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P1

Q.1

Linear algebra problem involving a rank-one matrix

$$Ax = b \quad -①$$

$A \Rightarrow m \times n$ matrix of rank 1
$b = m \times 1$ column vector

$$A = a_1 \cdot [1, \lambda_2, \lambda_3, --- \lambda_n] \quad -②$$

rank-1 decomposition: $A = a_1 \cdot v^T \quad -③$

putting value in eq^n ①

$$a_1 v^T x = b$$

so, b must be scalar multiple of a.

$$a_1 (v^T x) = \alpha a_1$$

$$v^T x = \alpha \quad -④$$

:- P can compute $\alpha = \frac{b_i}{a_{1i}} \quad (a_{1i} \neq 0)$

since $\boxed{v^T x = \alpha}$, P knows ~~typ~~ solutions.
P cannot find unique solution. but can find a solution $(v^T x = \alpha)$

↳ Q is wrong, P is right.

↳ P cannot find unique solution. P can find one correct solution.

Q.2 (P2)

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 0 \end{bmatrix} \quad -①$$

(a) SVD $\Rightarrow$

$$A = U \Sigma V^T$$

$U \to 2\times2$ orthogonal matrix

$\Sigma \to 2\times3$ diagonal matrix

$V^T \to 3\times3$ orthogonal matrix

$$A^TA \Rightarrow \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & -1 & 5 \\ -1 & 2 & 2 \\ 2 & 2 & 4 \end{bmatrix} \quad - ②$$

$|A - \lambda I| \Rightarrow 0$

$$\begin{vmatrix} 5-\lambda & -1 & 5 \\ -1 & 2-\lambda & 2 \\ 2 & 2 & 4-\lambda \end{vmatrix} = 0$$

On solving

$\lambda_1 \Rightarrow 6.6180$; $\lambda_2 = 4.3820$; $\lambda_3 = 0$

Singular values

$\sigma_1 = \sqrt{\lambda_1}$ $= \approx 2.5726$ $\quad \sigma_2 = \sqrt{\lambda_2}$ ≈ 2.0933 $\quad \sigma_3 = \sqrt{0}$ $= 0$

P3

$$\Sigma = \begin{bmatrix} 2.5726 & 0 \\ 0 & 2.0933 \\ 0 & 0 \end{bmatrix} - (3)$$

On solving λ values in V

$$V^T = \begin{bmatrix} -0.7394 & -0.1263 & -0.6613 \\ 0.5613 & -0.6575 & -0.5023 \\ -0.3714 & -0.7428 & 0.5571 \end{bmatrix} - (4)$$

$$\& \; U \Rightarrow \begin{bmatrix} -0.8507 & -0.5257 \\ -0.5257 & 0.8507 \end{bmatrix} - (5)$$

(b) Rank 1 Approximation

$$A_1 \Rightarrow \sigma_1 u_1 v_1^T$$

$$\Rightarrow 2.5726 \begin{bmatrix} -0.8507 & -0.5257 \\ -0.5257 & 0.8507 \end{bmatrix} \begin{bmatrix} -0.7394 & -0.1263 & -0.6613 \\ 0.5613 & -0.6575 & -0.5023 \\ -0.3714 & -0.7428 & 0.5571 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1.6180 & 0.2764 & 1.4472 \\ 1 & 0.1708 & 0.8944 \end{bmatrix} - (6)$$

2 (c)

SVD can help outliers by

- Reducing dimensionality :- It captures the most significant pattern in the data.
- Low-rank approximation

Q.③ (a) 8 x 4 data matrix

$$D \begin{bmatrix} 10000 & 120 & 25000 & 80000 \\ 9500 & 98 & 23000 & 60000 \\ 11000 & 140 & 30000 & 90000 \\ 12000 & 140 & 30000 & 100000 \\ 9000 & 100 & 24000 & 75000 \\ 8500 & 90 & 22000 & 55000 \\ 9000 & 105 & 260000 & 700000 \\ 100000 & 115 & 240000 & 780000 \end{bmatrix}$$

P5

AD001 = [10000, 120, 25000, 80000]

AD002 = [9500, 98, 23000, 60000]

Q3 (b)

Euclidean distance

$$d_{ij} = \sqrt{\sum_k (x_{ik} - x_{jk})^2}$$

Euclidean distance between AD001 & AD002

$$\Rightarrow \sqrt{500^2 + 22^2 + 2000^2 + 20000^2}$$

$\approx$

≈ 20105.98

Similarly if we follow the same approach for other data we can find the values.

d(AD001, AD003) ⇒ 11224.99

d(AD001, AD004) ⇒ 20712.32

d(AD001, AD005) ⇒ 5196.19

d(AD001, AD008) ⇒ 25224.01

d(AD001, AD009) ⇒ 10099.52

d(AD001, AD010) ⇒ 2236.07

d(AD002, AD003) ⇒ 30842.37

d(AD002, AD004) ⇒ 40684.79

d(AD002, AD005) ⇒ 15041.61

d(AD002, AD008) ⇒ 5196.16

d(AD002, AD009) ⇒ 10452.27

d(AD002, AD010) ⇒ 18034.70

d(AD003, AD004) ⇒ 10049.88

d(AD003, AD005) ⇒ 16278.87

d(AD003, AD008) ⇒ 35989.62

d(AD003, AD009) ⇒ 20493.93

d(AD003, AD010) ⇒ 13453.65

d(AD004, AD005) ⇒ 25884.39
d(AD004, AD008) ⇒ 45839.42
d(AD004, AD009) ⇒ 30413.83
d(AD004, AD010) ⇒ 22891.06
d(AD005, AD008) ⇒ 20105.97
d(AD005, AD009) ⇒ 5385.17
d(AD005, AD010) ⇒ 3162.31
d(AD008, AD009) ⇒ 15532.23
d(AD008, AD010) ⇒ 23135.48
d(AD009, AD010) ⇒ 8306.63

Final Data Matrix ⇒

	A001	A002	A003	A004	A005	A008	A009	A010
A001	0	20105.98	11224.99	20712.32	5196.19	25224.01	10099.52	2236.07
A002	20105.98	0	30842.37	40684.79	15041.61	5196.16	10452.27	18034.70
A003	11224.99	3084237	0	10049.38	16278.97	35989.62	20493.93	13453.65
A004	20712.32	40684.79	10049.88	0	25884.39	45839.42	30413.83	22891.06
A005	5196.19	15041.61	16278.87	25884.39	0	20105.97	5385.17	3162.31
A008	25224.01	5196.16	35989.62	45839.42	20105.97 ~~5385~~	0	1553223	23135.48
A009	10099.52	10452.27	20493.93	30413.83	5385.17	1553223	0	8306.63
A010	2236.07	18034.70	13453.65	2289.06	3162.31	23135.48	8306.63	0

↓

transpose matrix

Manhattan Distance

$d_{ij} \Rightarrow \sum_{k} |x_{ik} - x_{jk}|$

--- Continued.

(P7)

d(AD001, AD002)

$\Rightarrow |10000 - 9500| + |120 - 98| + |25000 - 23000| + |80000 - 60000|$

$\Rightarrow 22522$

similarly for the remainings

d(AD001, AD003) ⇒ 16020
d(AD001, AD004) ⇒ 27020
d(AD001, AD005) ⇒ 7020
d(AD001, AD008) ⇒ 29530
d(AD001, AD009) ⇒ 12015
d(AD001, AD010) ⇒ 3005

d(AD002, AD003) ⇒ 38542
d(AD002, AD004) ⇒ 49542
d(AD002, AD005) ⇒ 16502
d(AD002, AD008) ⇒ 7008
d(AD002, AD009) ⇒ 13507
d(AD002, AD010) ⇒ 19517

d(AD003, AD004) ⇒ 11000
d(AD003, AD005) ⇒ 23040
d(AD003, AD008) ⇒ 45550
d(AD003, AD009) ⇒ 26035
d(AD003, AD010) ⇒ 19025

d(AD004, AD005) ⇒ 34040
d(AD004, AD008) ⇒ 56550
d(AD004, AD009) ⇒ 37035
d(AD004, AD010) ⇒ 30025

d(AD005, AD008) ⇒ 22510
d(AD005, AD009) ⇒ 7005
d(AD005, AD010) ⇒ 4015

d(AD008, AD009) ⇒ 19515
d(AD008, AD010) ⇒ 26525

d(AD009, AD010) ⇒ 11010

So, the final matrix for Manhattan Distance

$$
\begin{array}{c|c}
 & \begin{matrix} AD001 & & & & & & & AD010 \end{matrix} \\
\begin{matrix} AD001 \\ \\ \\ \\ \\ \\ \\ AD010 \end{matrix} &
\begin{bmatrix}
0 & 22522 & 16020 & 27020 & 7020 & 29530 & 12015 & 3005 \\
22522 & 0 & 38542 & 49542 & 16502 & 7008 & 13507 & 19517 \\
16020 & 38542 & 0 & 11000 & 23040 & 45550 & 26035 & 19025 \\
27020 & 49542 & 11000 & 0 & 34040 & 56550 & 37035 & 30025 \\
7020 & 16502 & 23040 & 34040 & 0 & 22510 & 7005 & 4015 \\
29530 & 7008 & 45550 & 56550 & 22510 & 0 & 19515 & 26525 \\
12015 & 13507 & 26035 & 37035 & 7005 & 19515 & 0 & 11010 \\
3005 & 19517 & 19025 & 30025 & 4015 & 26525 & 11010 & 0
\end{bmatrix}
\end{array}
$$

Q.3)(c)

→ Euclidean distance is more influenced by large numeric differences due to squaring, whereas Manhattan Distance treats all dimensions linearly & is less sensitive to outliers

→ Manhattan distance can be easier to interpret & more robust for high-dimensional data.

Q.4

Gutter Problem:-

Let $x =$ slopes of length

total width $\Rightarrow 12m$

Horizontal projection $\Rightarrow x\cos\theta$

Vertical height $= x\sin\theta$

Horizontal base b/w the bends $\Rightarrow 12 - 2x\cos\theta$

Area of cross section

$$A \Rightarrow \text{height} \times \text{base}$$

$$A(x,\theta) \Rightarrow x\sin\theta \times (12 - 2x\cos\theta) \quad -(1)$$

$$A(x,\theta) \Rightarrow 12x\sin\theta - 2x^2\sin\theta\cos\theta \quad -(2)$$

$\Rightarrow$ Maximizing the area

:- w.r.t x

$$A_x = \frac{\partial A}{\partial x} \Rightarrow 12\sin\theta - 4x\sin\theta\cos\theta$$

Set $\frac{\partial A}{\partial x} = 0$

$$12\sin\theta = 4x\sin\theta\cos\theta$$

$$3 \Rightarrow x\cos\theta$$

$$x = \frac{3}{\cos(\theta)} \quad -(3)$$

Q.4 ... continued

$\therefore$ with respect to θ [substituting $x = \frac{3}{\cos\theta}$ in eqn 2

$$\Rightarrow \frac{3}{\cos\theta} \sin\theta \left(12 - 2\left(\frac{3}{\cos\theta}\right)\cos(\theta)\right)$$

$$\Rightarrow \frac{3\sin\theta}{\cos\theta}(12-6) \Rightarrow 18\tan(\theta)$$

To maximize A $\tan(\theta)$ should be maximum

but constraint to x

$$x = \frac{3}{\cos\theta} \leq 6 \quad (\text{since } 2x \leq 12)$$

so,

$$\theta = 60^\circ$$

$$x = \frac{3}{0.5} = 6 \leq 6$$

so, optimal angle $\theta = 60^\circ$

$$\boxed{x = 6}$$

Max area $\Rightarrow$

$$A = 18\tan(60^\circ)$$

$$\Rightarrow 18(\sqrt{3})$$

$$\approx 31.18\,m^2$$